

# Exercises learning task 19

2026 – 2027

**Exercise 11.1:** An article in *Concrete Research* [“Near Surface Characteristics of Concrete: Intrinsic Permeability” (1989, Vol. 41)] presented data on compressive strength  $x$  and intrinsic permeability  $y$  of various concrete mixes and cures. Summary quantities are  $n = 14$ ,  $\sum y_i = 572$ ,  $\sum y_i^2 = 23,530$ ,  $\sum x_i = 43$ ,  $\sum x_i^2 = 157.42$ , and  $\sum x_i y_i = 1697.80$ . Assume that the two variables are related according to the simple linear regression model.

- (a) Calculate the least squares estimates of the slope and intercept. Estimate  $\sigma^2$ . Graph the regression line.
- (b) Use the equation of the fitted line to predict what permeability would be observed when the compressive strength is  $x = 4.5$ .
- (c) Give a point estimate of the mean permeability when compressive strength is  $x = 3.3$ .
- (d) Suppose that the observed value of permeability at  $x = 3.3$  is  $y = 45.6$ . Calculate the value of the corresponding residual.

**Exercise 11.4:** An article in *Technometrics* by S. C. Narula and J. F. Wellington [“Prediction, Linear Regression, and a Minimum Sum of Relative Errors” (Vol. 19, 1977)] presents data on the selling price and annual taxes for 24 houses. The data are in the following table.

| Sale Price<br>/ 1000 | Taxes<br>(local, school, county)<br>/1000 | Sale Price<br>/ 1000 | Taxes<br>(local, school, county)<br>/1000 |
|----------------------|-------------------------------------------|----------------------|-------------------------------------------|
| 25.9                 | 4.9176                                    | 30.0                 | 5.0500                                    |
| 29.5                 | 5.0208                                    | 36.9                 | 8.2464                                    |
| 27.9                 | 4.5429                                    | 41.9                 | 6.6969                                    |
| 25.9                 | 4.5573                                    | 40.5                 | 7.7841                                    |
| 29.9                 | 5.0597                                    | 43.9                 | 9.0384                                    |
| 29.9                 | 3.8910                                    | 37.5                 | 5.9894                                    |
| 30.9                 | 5.8980                                    | 37.9                 | 7.5422                                    |
| 28.9                 | 5.6039                                    | 44.5                 | 8.7951                                    |
| 35.9                 | 5.8282                                    | 37.9                 | 6.0831                                    |
| 31.5                 | 5.3003                                    | 38.9                 | 8.3607                                    |
| 31.0                 | 6.2712                                    | 36.9                 | 8.1400                                    |
| 30.9                 | 5.9592                                    | 45.8                 | 9.1416                                    |

- (a) Assuming that a simple linear regression model is appropriate, obtain the least squares fit relating selling price to taxes paid. What is the estimate of  $\sigma^2$ ?
- (b) Find the mean selling price given that the taxes paid are  $x = 7.30$ .
- (c) Calculate the fitted value of  $y$  corresponding to  $x = 5.6039$ . Find the corresponding residual.

- (d) Calculate the fitted  $\hat{y}_i$  for each value of  $x_i$  used to fit the model. Then construct a graph of  $\hat{y}_i$  versus the corresponding observed value  $y_i$  and comment on what this plot would look like if the relationship between  $y$  and  $x$  was a deterministic (no random error) straight line. Does the plot actually obtained indicate that taxes paid is an effective regressor variable in predicting selling price?

**Exercise 11.6:** The following table presents the highway gasoline mileage performance and engine displacement for DaimlerChrysler vehicles for model year 2005 (U.S. Environmental Protection Agency).

| Carline              | Engine Displacement (inch <sup>3</sup> ) | MPG (highway) |
|----------------------|------------------------------------------|---------------|
| 300C/SRT-8           | 215                                      | 30.8          |
| CARAVAN 2WD          | 201                                      | 32.5          |
| CROSSFIRE ROADSTER   | 196                                      | 35.4          |
| DAKOTA PICKUP 2WD    | 226                                      | 28.1          |
| DAKOTA PICKUP 4WD    | 226                                      | 24.4          |
| DURANGO 2WD          | 348                                      | 24.1          |
| GRAND CHEROKEE 2WD   | 226                                      | 28.5          |
| GRAND CHEROKEE 4WD   | 348                                      | 24.2          |
| LIBERTY/CHEROKEE 2WD | 148                                      | 32.8          |
| LIBERTY/CHEROKEE 4WD | 226                                      | 28            |
| NEON/SRT-4/SX 2.0    | 122                                      | 41.3          |
| PACIFICA 2WD         | 215                                      | 30.0          |
| PACIFICA 4WD         | 215                                      | 28.2          |
| PT CRUISER           | 148                                      | 34.1          |
| RAM 1500 PICKUP 4WD  | 500                                      | 18.7          |
| RAM 1500 PICKUP 4WD  | 348                                      | 20.3          |
| SEBRING 4-DR         | 165                                      | 35.1          |
| STRATUS 4-DR         | 148                                      | 37.9          |
| TOWN & COUNTRY 2WD   | 148                                      | 33.8          |
| VIPER CONVERTIBLE    | 500                                      | 25.9          |
| WRANGLER/TJ 4WD      | 148                                      | 26.4          |

- (a) Fit a simple linear model relating highway miles per gallon ( $y$ ) to engine displacement ( $x$ ) in cubic inches using least squares.
- (b) Find an estimate of the mean highway gasoline mileage performance for a car with 175 cubic inches engine displacement.
- (c) Obtain the fitted value of  $y$  and the corresponding residual for a car, the Neon, with an engine displacement of 134 cubic inches.

**Exercise 11.8:** An article in the *Journal of Sound and Vibration* (Vol. 151, 1991, pp. 383-394) described a study investigating the relationship between noise exposure and hypertension. The following data are representative of those reported in the article

|     |    |    |    |    |    |    |    |    |    |    |
|-----|----|----|----|----|----|----|----|----|----|----|
| $y$ | 1  | 0  | 1  | 2  | 5  | 1  | 4  | 6  | 2  | 3  |
| $x$ | 60 | 63 | 65 | 70 | 70 | 70 | 80 | 90 | 80 | 80 |

  

|     |    |    |    |    |    |    |    |     |     |     |
|-----|----|----|----|----|----|----|----|-----|-----|-----|
| $y$ | 5  | 4  | 6  | 8  | 4  | 5  | 7  | 9   | 7   | 6   |
| $x$ | 85 | 89 | 90 | 90 | 90 | 90 | 94 | 100 | 100 | 100 |

**Exercise 11.18:** Show that in a simple linear regression model the point  $(\bar{x}, \bar{y})$  lies exactly on the least squares regression line.